M.E. MODEL PROJECT REVIEW PACK

AVA Evaluation

An Empirical Study of Conflict Resolution for Assertional State in Multi-Agent AI Systems

Agree · Verify · Abstain



A provisional, classroom-ready review aid — not an official KEC template

STATUS: proposal/review only. Pass 3 literature verification is incomplete; novelty is unresolved; the convergence condition is a target, not a proven theorem. No protocol implementation or formal thesis/paper drafting is authorized yet.

At a glance

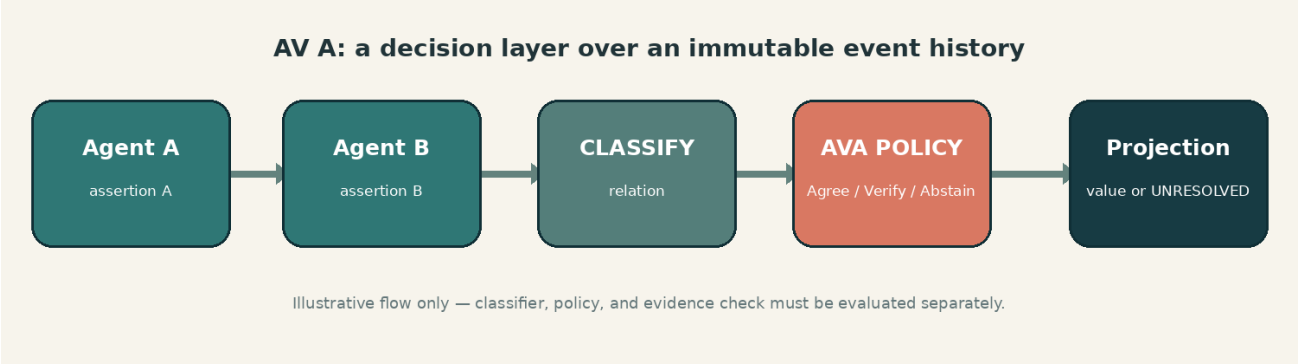
This pack turns the supplied finalized specification into a practical review presentation structure. The eight module titles below are provisional headings derived from that specification because no official college module list or review template was supplied.

LOCKED IN SPEC	PARTIAL LITERATURE EVIDENCE	OPEN / NOT AUTHORIZED

- Locked in the supplied spec: project name/title; same-key assertional-state focus; four relation labels; three policy actions; listed comparison arms; fault model boundaries; metrics and staged phases.
- Partial evidence: nearby studies already cover portions of semantic pair judgments, conflict-preserving state, provenance-aware resolution, abstention and asynchronous coordination. The comparisons below are scoped to the evidence saved so far.
- Open: exact novelty claim; WH-1/2/3 authoritative definitions; same-key schema and label rules; classifier/policy mapping; annotation capacity, sample size, budget and supervisor/college expectations.

60-second oral explanation

“When several AI agents update shared state, every replica can receive the same events while the meaning is still wrong or unsafe. AVA studies assertions about the same declared key: first classify the relationship, then decide whether to agree, verify with evidence, or preserve the disagreement as unresolved. I plan to compare deterministic, semantic, hybrid and structured methods using separate measures for false merges, unnecessary conflicts, calibration and cost, then test selected methods under delayed, duplicated and reordered delivery. SCP-MA is the test environment, not a claimed new protocol. The literature gap is still being checked, so this review presents a research proposal, not a confirmed novelty claim or completed implementation.”



Eight provisional review modules

1. Problem and background — Why shared meaning can fail even when replicas agree

2. Scope and state model — Separate additive facts from single-valued assertions

3. Relation classes — Classify the relationship before choosing a policy action

4. Agree / Verify / Abstain model — Keep policy actions distinct from terminal outcomes

5. Research questions and comparison arms — Compare strong baselines fairly, and separate classifier from policy

6. Dataset, annotation and metrics — Build a trustworthy gold set; report safety errors separately

7. SCP-MA experiment and convergence target — Use a minimal fault simulator only after the research gate

8. Risks, limitations and staged plan — Protect the research gate and keep the M.E. scope manageable

Provisional Module 1

Problem and background

Why shared meaning can fail even when replicas agree

Core idea

Several agents may write different statements about one shared fact. A replicated log can deliver every event to every replica and still leave the application with an unsafe or misleading meaning. AVA studies the decision made after same-key disagreement is detected; it is not a general consensus protocol or universal truth checker.

What to say / show

- Motivation: mechanical convergence is not semantic correctness.
- Question: which assertion relationships can be safely reconciled, and when should the system verify or abstain?
- Falsifiable stance: deterministic methods may beat semantic models; either result is valid.



Animal illustration is a teaching analogy only — not evidence or a model of the protocol.

Short talk track

"I am not claiming that AI must be better. I want to measure when each method can safely resolve a disagreement and when it should leave the state unresolved."

Likely review question

Ask: Is this a real systems problem? Answer: yes, as a bounded empirical problem; the size and originality of the gap still need verified literature comparison.

Provisional Module 2

Scope and state model

Separate additive facts from single-valued assertions

Core idea

The proposal separates state into additive state A and assertional state Q. Additive observations such as logs can be accumulated with standard CRDT structures. The focal case is a state key intended to have one value at a time, where competing values may conflict. An assertion records key, value, source agent, causal metadata such as a vector clock, and logical timestamp.

What to say / show

- In scope: pairs concerning the same declared key; cooperative agents; stale data; concurrent writes; delay, duplication, reordering, disconnection and reconnection.
- Out of scope: Byzantine agents, permanent partitions, general open-domain contradiction, cross-key intent conflicts and organizational governance.
- Key identity/cardinality and typed value schema must be specified before examples are labeled.

Short talk track

“The key is the anchor. If two values refer to different roles or time scopes, they may not be a conflict at all. The dataset must make that scope explicit.”

Likely review question

Ask: Why not solve every conflict? Answer: cross-key invalid combinations and adversarial behavior are separate research problems and are explicitly excluded.

Provisional Module 3

Relation classes

Classify the relationship before choosing a policy action

Core idea

The specification names four pair classes. Paraphrase means the same fact in different wording. Compatible means both statements can be true together. Soft conflict is a genuine difference that a stated policy or temporal context could resolve. Hard conflict means mutually exclusive assertions that are unsafe to auto-merge. The labels concern the pair's relationship, not the system's action.

What to say / show

- Paraphrase: "PostgreSQL is selected" / "The project uses PostgreSQL."
- Compatible: two statements can both hold under clearly declared distinct roles/scopes.
- Soft conflict: API v1 is in use while migration to v2 is underway, if the temporal transition is represented.
- Hard conflict: database=PostgreSQL versus database=DynamoDB for the same role and time.



Animal illustration is a teaching analogy only — not evidence or a model of the protocol.

Short talk track

"First I label what relation the pair has. Only after that do I ask whether a policy should agree, verify, or abstain."

Likely review question

Design risk: the provided PostgreSQL/DynamoDB compatible example describes different database roles, so it may violate same-key scope unless the key includes role/context.

Provisional Module 4

Agree / Verify / Abstain model

Keep policy actions distinct from terminal outcomes

Core idea

AVA has three policy actions: Agree, Verify, Abstain. Verify is intermediate. It consults additional evidence such as provenance, timestamps, causal metadata, or deterministic rules. If the evidence supports a safe resolution it terminates as verified-agree; otherwise it terminates in abstention. Abstention preserves the competing assertions as UNRESOLVED rather than deleting or guessing.

What to say / show

- Direct Agree: sufficient support for immediate reconciliation.
- Verify: do not merge yet; run an evidence check.
- Abstain: preserve conflict and defer resolution.
- In convergence-proof experiments, Verify must be a deterministic pure function of shared canonical inputs; a live stochastic model call is not convergence-safe.



Animal illustration is a teaching analogy only — not evidence or a model of the protocol.

Short talk track

"Verify is not just choosing the newest write. It is a second-stage evidence check. If evidence is not enough, the system keeps both claims unresolved."

Likely review question

Ask: How is Verify different from B1? Answer: B1 is a winner-selection rule; Verify asks whether additional deterministic evidence justifies a safe decision.

Provisional Module 5

Research questions and comparison arms

Compare strong baselines fairly, and separate classifier from policy

Core idea

The primary question is whether decision mechanisms can reliably classify same-key assertion pairs and determine when to agree, verify, or abstain. The specification lists seven arms: deterministic winner selection; embeddings; NLI; a hybrid SCS formulation; LLM-as-a-Judge; Jev (System-One decision model); and conflict-preserving abstention.

What to say / show

- RQ1–RQ2: relation classification and semantic-versus-deterministic performance.
- RQ3–RQ4: confidence-aware policy, calibration, latency and cost.
- RQ5–RQ6: asynchronous behavior and scaling. RQ7: trust-weighting ablation.
- Fairness issue: B1 and B7 are policies, while B2–B6 are primarily classifiers. Report classification separately from final resolution outcomes, with shared inputs and explicit mappings.

Short talk track

"I will not compare a classifier and a safety policy as if they do the same job. Classification quality and final resolution safety will be reported separately."

Likely review question

Open choice: determine whether the main thesis is a classifier benchmark, a policy comparison, or a narrow staged combination. Keep seven-arm breadth provisional until scope and resources are checked.

Provisional Module 6

Dataset, annotation and metrics

Build a trustworthy gold set; report safety errors separately

Core idea

Each record is intended to include assertion pair, declared key, agents, context, temporal context, evidence, gold relation label and difficulty. The proposed construction has controlled synthetic examples, LLM-generated realistic examples, and real traces only where lawful and practical. Ground truth requires multiple human annotators, agreement measurement and adjudication; an LLM alone is not a gold-label authority.

What to say / show

- Classification: accuracy, precision, recall, macro/per-class F1.
- Safety: FMR counts incorrect Agree outcomes on true conflicts; FCR counts failures to safely resolve true paraphrase/compatible cases. Unnecessary abstention is a named subset of FCR and is intentionally reported separately too.
- Calibration: ECE, Brier, reliability diagrams, risk–coverage, selective accuracy. Cost: latency, tokens/API calls, compute, throughput.
- Pre-register annotation guide, key schema, split strategy, leakage controls and sample-size rationale before benchmark implementation.

Short talk track

“A false merge can be dangerous, so I report it separately from needless abstention. I will not hide either inside one score.”

Likely review question

Open decisions: annotation capacity, sample size/power, adjudication protocol, domain balance and whether context/provenance is visible to each classifier.

Provisional Module 7

SCP-MA experiment and convergence target

Use a minimal fault simulator only after the research gate

Core idea

SCP-MA is the evaluation environment, not a claimed novel protocol. It models concurrent cooperative agents, immutable events, causal metadata, asynchronous delivery faults and eventual reconciliation. State is projected from an event history; semantic conflict does not trigger destructive rollback. The target property says that replicas with the same valid event set should compute the same deterministic projection after canonicalization.

What to say / show

- Validate event-set convergence in a semantics-free simulator before adding semantic decision methods.
- Inject delay, duplication, reordering, temporary disconnection and reconnection. Measure convergence rate/latency, divergence, messages/bytes, recovery and duplicate handling.
- For convergence-critical arms, classification/Verify must be deterministic or use decision caching/replay. Model-based Verify is a separate non-convergence-safe condition.
- Do not describe the target as a proved theorem: determinism, canonicalization and exact projected state remain obligations.



Animal illustration is a teaching analogy only — not evidence or a model of the protocol.

Short talk track

"The simulator tests whether delivery faults change the result. First I would prove the event mechanics with no semantic model; then add only methods that meet each arm's stated determinism conditions."

Likely review question

Ask: Is SCP-MA the novelty? Answer: no; it is the experimental scaffold. Any novelty claim must come from verified research question and evidence, not the simulator name.

Provisional Module 8

Risks, limitations and staged plan

Protect the research gate and keep the M.E. scope manageable

Core idea

The project remains at the literature and design gate. A source-wide primary-text pass across five required papers is now recorded, but overall Pass 3 and level-P verification remain incomplete; wider prior-art/NLI checks and the exact WH-1/2/3 definitions are unresolved, and convergence is a target rather than a proof. No protocol implementation or formal thesis/paper drafting is authorized yet. This pack is for review discussion only.

What to say / show

- Gate 1: integrate the five source-wide audits, finish residual closest-threat/NLI reconciliation, and actively challenge the candidate gap.
- Gate 2: settle key schema, label rules, policy/classifier separation, evidence access, and safe Verify semantics.
- Gate 3: check college scope, annotation feasibility, compute/API budget and supervisor expectations; these facts are not supplied.
- If the question survives: prioritize a small reproducible benchmark and safety–utility analysis; keep the simulator minimal and scaling as stretch work. If not, narrow, reframe, or abandon rather than retrofitting novelty.

Short talk track

“My next step is not coding. It is to finish the literature attack and resolve the design decisions. If the gap does not survive, I will narrow or change direction instead of claiming novelty.”

Likely review question

Unverified facts not to invent: official KEC module names/template, supervisor requirements, deadlines, annotation pool, budget, results, and implementation status.

How the seven arms fit together

The supplied specification lists seven arms, but they do not all operate at the same layer. Treat the table as the planned comparison, not evidence that a fair design is already settled.

Arm	Role in the study	Fair-comparison note
B1 deterministic argmax(timestamp, agent_id)	Winner-selection baseline	Resolution policy; not itself a four-class classifier.
B2 embeddings	Semantic similarity signal	Similarity is not logical compatibility; evaluate empirically.
B3 NLI	Entailment / contradiction / neutral signal	Specify pair/context input consistently.
B4 hybrid SCS	Cosine-distance + distance term	Weights/formula are experimental choices, not assumed optimal.
B5 LLM-as-a-Judge	Generative pair judgment	Fix model/version/prompt/temp; repeat and report variance.
B6 Jev (System-One decision model)	Structured decision questions	Compare calibration, accuracy, cost with same evidence access.
B7 conflict-preserving abstention	Safety baseline: preserve as UNRESOLVED	Policy baseline; measure resolution coverage and safety separately.

Metrics in plain language

False-Merge Rate (FMR)

Among genuinely conflicting cases, the fraction incorrectly ended in Agree, whether direct or after Verify. Report this separately because an unsafe merge can be costly.

False-Conflict Rate (FCR)

Among genuine Paraphrase + Compatible cases, the fraction not safely resolved. The specification includes unnecessary abstentions here.

Unnecessary-abstention rate

The subset of FCR cases that reached Verify and then abstained despite resolvable ground truth. It overlaps FCR intentionally; these metrics are not additive.

Calibration and coverage

For probabilistic methods, check whether confidence matches observed correctness, and plot accuracy as the system abstains more or less.

Distributed behavior

Measure whether replicas converge to the defined projection, how long it takes, divergence during faults, message/byte costs, duplicate handling and recovery.

Literature: useful comparisons, carefully bounded

These are partial/qualified comparisons from the project's saved source audit, not a final systematic review. The aim is to show examiners that nearby work is being taken seriously, while avoiding a claim that the field has no prior work.

- MELD — A Protocol for Merging Knowledge Across Distributed Agentic Memories — Closest pair-taxonomy overlap; 104 balanced pairs deliberately use distinct claim keys; status convergence is not semantic-view convergence.
- StateFuse — Deterministic Conflict-Preserving Memory for Multi-Agent Systems — Conflict objects, provenance, immutable history, deterministic projection and abstention overlap; reviewed sections do not establish identical task.
- TOKI — A Bitemporal Operator Algebra for Contradiction Resolution in LLM-Agent Persistent Memory — Temporal records, provenance, typed operators, evidence/confirmation policies; section-level evidence in project notes.
- SubtleMemory — A Benchmark for Fine-Grained Relational Memory Discrimination — 1,522 downstream QA instances over 10 long histories; relational memory task, not isolated assertion-pair classification.
- S-Bus — Automatic Read-Set Reconstruction for Multi-Agent LLM State Coordination — Structural concurrency control; 45/45 OCC rejections in a specified experiment; not semantic classification.
- Semantic Consensus Framework (SCF) — Reported 27.9% precision, 65.2% recall, F1 39.0 for its workflow-intent conflict task; not same-key assertion pairs.
- CodeCRDT — Observation-Driven Coordination for Multi-Agent LLM Code Generation — 5–10% semantic-conflict estimate from manual inspection of 60/600 runs; preliminary, not general rate.
- Governed Shared Memory for Multi-Agent LLM Systems — Same-subject/property supersession study; 90/90 admitted conflicts vs 98/200 overall due to admission effects, as recorded in project notes.
- Reliable Post-Retrieval Assembly for Agent Memory — +10.8/+21 points are whole-pipeline result; executor-only comparison +2/0 and LongMemEval 26/45 vs 29/45, p=.45.
- ECon — On the Detection and Resolution of Evidence Conflicts — Question-conditioned evidence-pair conflict task; adjacent NLI baseline, not replicated keyed state.
- RefNLI — Reference ambiguity and context mismatch complicate contradiction labels.
- Selective Prediction for NLP — Selective prediction/abstention predates AVA; not direct distributed same-key experiment.

Numbers to state only with their scope

- MELD: the 104-pair balanced diagnostic set intentionally uses distinct claim keys; its reported pair-classification results therefore do not directly establish performance on a strict same-key-only dataset.
- CodeCRDT: the 5–10% semantic-conflict estimate is based on manual inspection of 60 of 600 runs, a preliminary sample rather than a validated general rate.
- SCF: reported Table 2 values are 27.9% precision, 65.2% recall and F1 39.0 for its workflow-intent conflict task; 65.2% is not accuracy.
- Deterministic post-retrieval assembly: +10.8 percentage points on average / +21 at context length 262K compares whole pipelines. The executor-only comparison is +2 average / 0 at 262K; on LongMemEval, 26/45 vs 29/45 with p=.45.
- SubtleMemory v3: 1,522 downstream QA instances across 10 persona-level histories and 1,090 relation-controlled variant sets. It tests long-history relational memory discrimination, not isolated assertion-pair classification.

Open decisions and review-ready answers

What exactly is the key?

Define cardinality, type, scope, role and temporal granularity. A compatible pair must either share one key by design or be clearly identified as a different-key/control case.

Are class labels independent of policy?

Yes in the proposed design: relation class is a gold annotation; Agree/Verify/Abstain is an action. Specify how each arm maps predictions/evidence to actions.

What does Verify inspect?

Specify which deterministic evidence is available and what threshold/rule is sufficient. For convergence experiments, both replicas with identical canonical evidence must produce identical Verify outcomes.

What is the exact thesis novelty?

Not established. The safest candidate distinction is a narrowly scoped empirical question about evidence-based Verify after same-key classification under asynchronous delivery, but StateFuse/TOKI/MELD and other work require full comparison.

What are WH-1, WH-2, WH-3?

The supplied finalized specification names but does not define them. Do not present any agent-written paraphrase as the owner’s exact hypotheses.

How large is the work?

Seven arms + a human-labeled dataset + calibration + a simulator + scaling is ambitious. Keep the core benchmark/safety results first; treat simulator depth and scaling as staged/stretch scope subject to requirements and feasibility.

What has been implemented?

Nothing in the thesis protocol is authorized or claimed implemented. The current work is planning and literature/design reconciliation.

Staged plan (no dates invented)

1. Stage A — Literature gate: source-wide/appendix verification, NLI/contradiction review, active attempt to disprove the candidate gap; record exact versions and review scope.
2. Stage B — Design preregistration: key schema/cardinality; label guide; evidence visibility; classifier/policy separation; split/leakage controls; sample-size and resource plan.
3. Stage C — Core benchmark, only if the gate and approval conditions are met: controlled dataset and human annotation, deterministic baseline, selected semantic methods, safety/calibration/cost reporting.
4. Stage D — Minimal distributed simulator: first test event canonicalization and deterministic projection without semantic methods; then integrate only conditions compatible with the convergence claim.
5. Stage E — Scope decision and writing: expand, narrow or reframe based on measured results and college/supervisor requirements; only proceed to implementation or formal thesis drafting after the project’s gate is satisfied.

Sources and status note

Primary source for the locked project choices: “AVA Evaluation — Finalized Thesis Specification” supplied by Selvin. Source links above point to public versions where available; literature summaries and metrics are quoted only with the scope limitations preserved in the saved project audit. Please verify versions/citations against full texts before formal submission.

This review pack is not an official college format, an approved thesis, a completed literature review, an implementation plan authorization, or a report of experimental results. It is an explanatory proposal aid made from the supplied specification and qualified evidence in conversation history.

Short closing pitch

“AVA asks a practical question: when agents disagree about the same state key, what can be reconciled safely, what needs evidence, and what should remain unresolved? I will compare methods with explicit safety and usefulness measures, without assuming that more intelligent models perform better. The current contribution is a testable proposal; the novelty and convergence claims stay open until their gates are met.”

Review focus: why shared state matters in an AI-agent team

This is a hypothetical teaching scenario using the six labels exactly as supplied: **Grok bot, GPT dots, Claude duet, Harkpro, Wajo Fo, Wabi**. These are labels only. No real capability, product identity, integration, role, or performance is being asserted.

Assume the six independently operating agents each emit a candidate value for the same explicitly declared key: **service.demo-api.primary_db**. The key has type “database enum,” exactly-one cardinality, scope “demo-api,” and a shared current-release interval. Three hypothetical writes say PostgreSQL and three say DynamoDB. The example is constructed only to make the method easy to explain; it is not a real system trace or an experimental result.

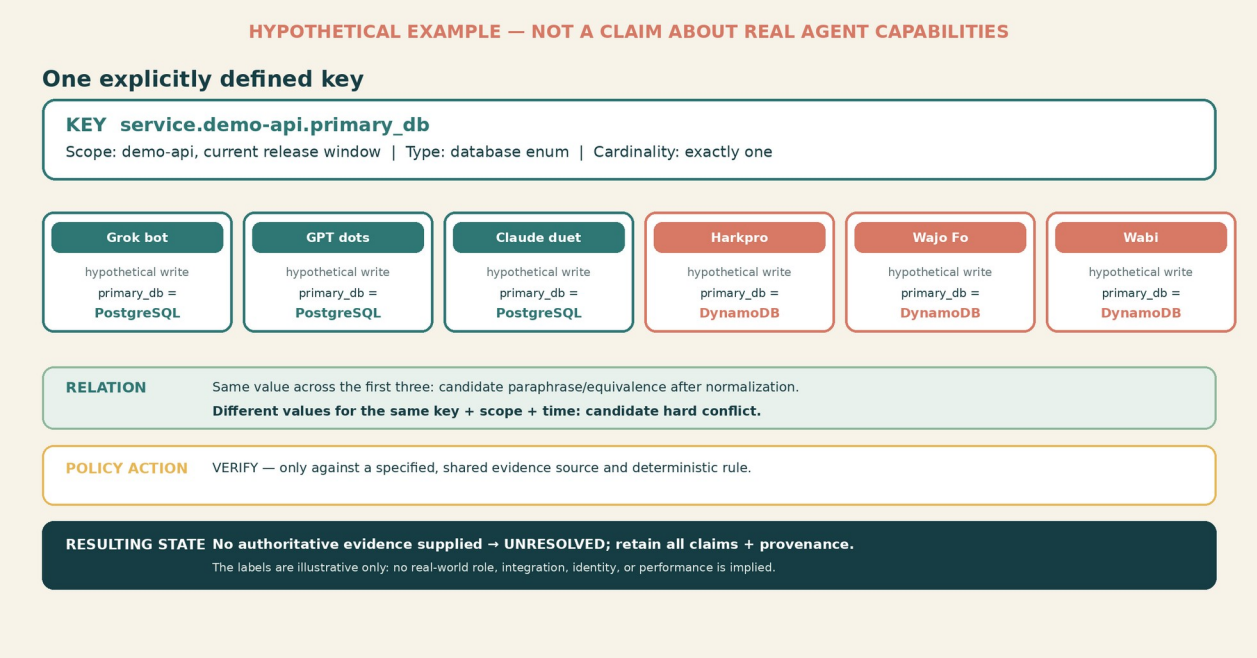


Figure: provenance → relation → policy action → resulting state. Teaching illustration only; not evidence or a model of actual named agents.

How to explain the four separate questions

- **1. Provenance — who supplied which claim?** Keep each assertion attached to its hypothetical label and event metadata. A label alone is not a reliability score.
- **2. Relation — can the claims coexist?** With the key, role, interval and cardinality held fixed, equal normalized values can be treated as equivalent candidates; PostgreSQL versus DynamoDB is a hard-conflict candidate. If those scope fields differ, do not call it a same-key conflict.
- **3. Policy action — what should the system do?** The example chooses Verify as a proposed action because claims disagree. Verify is not a result; its evidence source and decision rule must be specified before any conclusion.
- **4. Result — what state is published?** No shared authoritative evidence is supplied in the example, so the safe demonstration outcome is UNRESOLVED with both candidate values and provenance retained. It does not guess which database is correct.

Extra review points to make the proposal stronger

1. State the key contract first. Name the key type, value domain, scope, temporal interval and cardinality. Make every pair either genuinely same-key or explicitly a control. This prevents a role mismatch from being mislabeled as a conflict.

2. Separate four layers. Keep agent/event provenance, gold relation label, policy action (Agree/Verify/Abstain), and final state (resolved value or UNRESOLVED) as separate fields. This makes errors diagnosable.

3. Give Verify a bounded job. Name the permitted evidence, freshness rule, conflict tie behavior and deterministic decision procedure. If evidence is missing or contradictory, preserve UNRESOLVED rather than inventing certainty.

4. Make comparisons fair. Use the same pair, context, evidence, prompt budget and evaluation split across arms where applicable. Report classifier quality apart from policy outcomes; compare the deterministic and preserve/abstain baselines directly.

5. Measure safety and usefulness separately. Report false merges on real conflicts, false conflicts on resolvable pairs, unnecessary abstentions, class-wise scores, calibration, latency and cost. Avoid a single score that hides safety trade-offs.

6. Keep the scope feasible. Seven methods plus annotation, calibration and distributed simulation is ambitious for an M.E. First decide the research gate and a small reproducible core; make simulator scale-up conditional, not assumed.

Questions a reviewer may ask — concise answers

Why use a team of agents rather than one assistant?

The team is an illustrative setting for independently produced updates to shared state. The research question is about the state decision, not a claim that any named agent is connected or has a particular role.

Why not always take the newest write?

Arrival time may reflect network order rather than truth or validity. Newest-wins is a useful baseline to test, not a default truth rule.

Does Verify guarantee correctness?

No. It can only use specified evidence and rules. Without adequate evidence, it should abstain or leave the key unresolved.

Does this prove all replicas converge?

No. Convergence is a target condition. The event set, canonicalization, resolver determinism and exact meaning of projected state still need definition and verification.

Is the novelty established?

No. Pass 3 remains incomplete and prior work overlaps with several parts of the proposal. The candidate gap must survive source-wide review and active disconfirmation.

What do the six agent names mean?

They are user-supplied hypothetical labels for the teaching scenario only. The example assigns no actual identity, capability, integration, role or performance to them.

Monday 26 October review: presentation plan

Use this as a practical rehearsal checklist for Monday. The review start time and official college module labels were not supplied, so this pack keeps its eight module headings provisional and does not invent a schedule.

Opening talk track (about 90 seconds)

> “My proposal studies a narrow shared-state problem in multi-agent AI: several agents may update one key, but receiving all events does not guarantee that the published meaning is safe. I define the key and scope first, classify the assertion relationship, then compare Agree, Verify and Abstain policies. For example, six hypothetical agent labels can write different values for one explicitly defined database key; if evidence does not settle the disagreement, the result stays UNRESOLVED with provenance preserved. I will measure unsafe merges, missed safe resolutions, unnecessary abstention, calibration and cost, and only then consider a minimal asynchronous simulator. Existing work overlaps with important parts of this proposal, so novelty is not established; the Pass 3 literature gate is still open. Today I’m seeking feedback on the key schema, label definitions and feasible scope—not presenting results or a completed implementation.”

Before you present

- Open the PDF slide deck once; confirm the slide order and that fonts/diagrams display on the presentation device.
- Keep the editable PPTX and PDF together; bring the report PDF as a reference copy and the editable report only if asked.
- Rehearse the 90-second opener and the one-key example; point to provenance, relation, policy action, then result in that order.
- Say “hypothetical example” before using the six agent labels. Do not describe them as connected products or assign them real functions.
- State the gate honestly: literature review incomplete, novelty unresolved, convergence unproved, no protocol implementation or formal thesis drafting.
- Ask which college module headings/template should replace the provisional eight-part structure, and what the review expects next.
- Write down the supervisor/reviewer’s requested changes and any date they specify; do not assume a milestone or approval from discussion alone.

Three useful feedback questions for the review

- Is the same-key contract (key, value type, role, time scope, cardinality) precise enough for the first review?
- Should the M.E. prioritize relation classification and safety metrics first, with distributed simulation kept minimal or staged?
- What required literature/design evidence must be completed before the project may move beyond the proposal gate?

Monday-ready additions prepared 22 October 2026. The scenario, agent labels and animal references are teaching aids only; they are not source evidence, system observations or implementation claims.

Appendix A — Current Primary-Source Audit Notes

Prepared 22 October 2026 · evidence checkpoint only · not a final literature verdict

How to read this appendix

These notes combine a source-wide primary-text pass for Always-On Agents, StateFuse, TOKI, S-Bus and SCF with selected-section checks of other papers. Exact scope and source limitations are recorded in the Pass 3 matrix; these notes are not independent reproductions, an exhaustive literature census, or an adjudicated novelty decision. Full-text reading does not by itself satisfy the project's level-P standard. Wider threat/NLI checks and the novelty gate remain open. No implementation or formal thesis claim is authorized by this appendix.

Working question: does AVA's proposed same-key assertion pipeline—classify a pair, then use evidence-based Verify—add anything beyond fixed-winner and preserve-unresolved policies when delivery is asynchronous? This remains a candidate question, not a demonstrated gap. Compare semantic-state convergence separately from event delivery, status convergence, and downstream retrieval.

Closest systems and memory-policy comparisons

MELD — A Protocol for Merging Knowledge Across Distributed Agentic Memories (arXiv:2608.16357v1)

Primary source — https://arxiv.org/html/2608.16357v1

The 104-pair benchmark deliberately uses distinct claim keys, so it is not a strict same-key pair test. Its status CRDT converges after healing, but the merge graph/live claim set/retrieval view can remain order-sensitive: Appendix M reports 177 joint graph fingerprints across 200 orderings and different live sets in 112/200. Earlier checked findings: 26 seeds × four classes; classifier macro-F1 .845 (95% CI .766–.912), AUC .968 and FMR 1/78. This is close relation-label/convergence prior art, but does not settle AVA's same-key policy comparison. Reading remains selected, not source-wide.

MemTX — Transactional Belief Commit for Stateful Agent Memory (arXiv:2607.23929v2)

Primary source — https://arxiv.org/html/2607.23929v2

Evidence-, permission-, provenance- and validity-bearing records are staged under snapshot isolation and admitted through validate-and-commit; a rule-based semantic-conflict check handles the same entity/attribute slot. The paper reports a 90-case main suite across five backbones and

bounded checking over about 5.5 million states. Cases use 1–3 agents (median 2) and interleaved turns over one shared store. This is close semantic commit/conflict work, not evidence of decentralized asynchronous replica convergence.

TARL — Transaction-Aware Reliable Ledgers for Executable Memory Management (arXiv:2608.03699v5)

Primary source — https://arxiv.org/html/2608.03699v5

Its five actions are append, noop, revise, reject_conflict and defer_verify; Pending and Rejected are distinct destinations. TARL-Mem contains 5,422 cases and includes exact gold next-state behavior, state recovery, conflict preservation, temporal handling and sequential rollout. This directly overlaps deferral vs rejection and fine-grained transaction policy; the checked material does not evaluate asynchronous distributed replication or convergence.

LatticeMind — A Conflict-Aware Memory Primitive for Multi-Agent Systems (arXiv:2608.08236v1)

Primary source — https://arxiv.org/html/2608.08236v1

Routes keyed claims and classifies semantic conflicts as Credibility or Coordination before reconciliation. On 75 label-blind ConflictBank examples with Qwen3-Max, it reports 73/75 (0.97), versus 0.61 for the strongest aggregation baseline. The benchmark is small and the inspected evaluation uses centralized/item-state handling; no asynchronous writer/convergence result was found. Strong overlap for classify-then-reconcile, not proof of AVA-specific preemption.

StateFuse — Deterministic Conflict-Preserving Memory for Multi-Agent Systems (arXiv:2607.05844v1)

Primary source — https://arxiv.org/html/2607.05844v1

Uses replicated CRDT state with explicit conflicts, provenance/correction handles and projection-time selection or abstention without mutating the base state. On the 282-question MemoryAgentBench slice, Table 3 reports separate operating settings: conservative abstention has 64.9% accuracy, 100% conflict recall and 2.1% false certainty; the aggressive StateFuse setting has 97.5% accuracy, 100% conflict recall and 2.5% false certainty. These are not contradictory results. In a separate controlled 50-task loop with a uniform verification budget, the conservative StateFuse surface reports 80% pre-verification and 100% post-verification success, false certainty falling from 20% to 0%, 2.4 average tool calls and 0.00 average action cost. In Table 4, conservative flat baselines also go from 80% to 100% post-Verify success, while collapsed goes from 40% to 60% and retains 40% post-Verify false confidence. This controlled 50-task loop is not a decentralized asynchronous-convergence evaluation. Main text and Technical Appendix A.1–A.5 were read; per-question outputs and some implementation details are not present in the source page.

TOKI — A Bitemporal Operator Algebra for Contradiction Resolution in LLM-Agent Persistent Memory (arXiv:2606.06240v1)

Primary source — https://arxiv.org/html/2606.06240v1

The contradiction rule requires the same subject and predicate, different object, and overlapping valid time. Four operators (LWW, Evidence, AwaitConfirm and PerRule) cover read-committed

through serializable policies; they stamp a winner and merged-provenance audit row. Table 19 reports +0.86 on LoCoMo audit rows (n=50, 95% CI [.76,.94]); the 1,444-question answerable-pool ablation is .540 with memory vs .048 without. Table 24 is a single-node PostgreSQL test, not distributed deployment evidence. Main text and Appendices A–H were read; supplied HTML omits references and raw experiment files.

CoAgent — Concurrency Control for Multi-Agent Systems (arXiv:2606.15376v1)

Primary source — https://arxiv.org/html/2606.15376v1

An LLM judges whether a concurrent write invalidates its plan and repairs affected actions. Serializability at quiescence depends on stated protocol assumptions, including correct relevance judgments; authors report 5% misjudgment in their tested setting. This is plan-validity repair, not adjudicating which same-key assertion is true.

Governed Shared Memory for Multi-Agent LLM Systems (arXiv:2606.24535v1)

Primary source — https://arxiv.org/html/2606.24535v1

Reports post-commit asynchronous contradiction supersession in 90/90 runs conditional on both writes being admitted; overall it is 98/200 because a synchronous dedup gate rejected 206/400 writes before the asynchronous detector saw them. This is the closest direct asynchronous shared-memory evidence found and shows why admission effects belong in end-to-end metrics. It does not establish AVA's verifier novelty.

S-Bus — Automatic Read-Set Reconstruction for Multi-Agent LLM State Coordination (arXiv:2605.17076v2)

Primary source — https://arxiv.org/html/2605.17076v2

Observable-Read Isolation covers HTTP-observable reads only: Table I reports 26.1% R_obs coverage and 73.9% hidden reads. Table X reports 0/638 divergent commits with ORI vs 590/639 without across 80 trials. Formal proofs cover 687 TLAPS obligations modulo one retained axiom; this is structural stale-read protection, not semantic assertion verification. Main text and Appendix A were read; references are absent from the supplied page.

Semantic Consensus / SCF — Process-Aware Conflict Detection and Resolution for Enterprise Multi-Agent LLM Systems (arXiv:2604.16339v1)

Primary source — https://arxiv.org/html/2604.16339v1

Detects workflow-intent conflicts before execution, not replicated claim truth. Across 600 simulated runs it reports 27.9% precision, 65.2% recall, 39.0% F1 and 100% workflow completion; the setting covers 4–6 agents. The +10.8/+21 percentage-point result in the related evidence is whole-pipeline, not isolated policy gain; on LongMemEval the isolated policy comparison is 26/45 vs 29/45, p=.45. Live distributed convergence is not shown. Main text §§1–8 and four tables were reviewed; no appendices are listed, but bibliography entries are absent from the retrieved HTML.

Always-On Agents survey (arXiv:2606.30306v1)

Primary source — https://arxiv.org/html/2606.30306v1

A scoped, non-exhaustive map of 435 works. Table 18's AOEP pilot covers nine fault patterns and seven systems; the governed reducer passes 15/15 obligations and 41/41 negative checks, while no-memory passes 0/15 and 41/41, exposing a vacuous-pass issue. Appendix A reports coder agreement on a 236-work sample (lifecycle .82, κ=.58; state .74, κ=.44; rollback agreement on 11/19 co-flagged works). Read HTML through §12, Appendix A and references; the governance-envelope ablation remains unrun and cited-study data are not all included.

CodeCRDT — Observation-Driven Coordination for Multi-Agent LLM Code Generation (arXiv:2510.18893v1)

Primary source — https://arxiv.org/html/2510.18893v1

The often-cited 5–10% semantic-conflict rate is a preliminary manual estimate from 60/600 runs, not a fully measured population result. It distinguishes mechanical convergence from semantic quality but is not a same-key assertion benchmark.

SubtleMemory v3 (arXiv:2606.05761v3)

Primary source — https://arxiv.org/html/2606.05761v3

Relational long-history downstream QA with complementary/nuanced/contradictory cases. It is not isolated same-key pair classification or asynchronous state convergence. It narrows broad relation-sensitive memory-evaluation claims, not the exact AVA systems question.

NLI, evidence verification and selective-prediction stress tests

ECon — On the Detection and Resolution of Evidence Conflicts (arXiv:2410.04068v1)

Primary source — https://arxiv.org/html/2410.04068v1

Studies question-conditioned answer/factoid evidence-pair conflicts. Its reported pattern is that NLI and LLM detectors can have high precision while weaker models miss conflicts; resolution can be unjustified or rely on the model's prior knowledge. This is verifier stress evidence, not a concurrent shared-state protocol. Full result-table reconciliation remains open.

REF-NLI — On Reference (In-)Determinacy in Natural Language Inference (Findings of NAACL 2025)

Primary source — https://aclanthology.org/2025.findings-naacl.450/

Its 1,143 claim/evidence pairs include ambiguous references and context mismatch. This warns that a verifier may call claims contradictory when their referents or contexts do not align. It is an annotation/robustness resource, not distributed state resolution.

ANLI v2 — Adversarial NLI (arXiv:1910.14599v2)

Primary source — https://arxiv.org/html/1910.14599v2

Three adversarial rounds contain roughly 19k, 47k and 103k examples. Dev/test examples were human-verified; the paper says training examples were not all individually verified. Useful for stress-testing NLI components, not evidence of memory-policy novelty.

Varshney, Mishra & Baral — Investigating Selective Prediction Approaches Across Several Tasks in IID, OOD, and Adversarial Settings (arXiv:2203.00211v1)

Primary source — https://arxiv.org/abs/2203.00211

The supplied eight-page PDF was read. Across 17 datasets spanning NLI, duplicate detection and QA, it evaluates risk–coverage selective prediction. No method consistently and considerably beats MaxProb across IID, OOD and adversarial settings; performance varies by task and degrades on adversarial cases. Abstention/confidence alone is not evidence verification and needs task-matched evaluation.

What the evidence supports — and does not

- Unsafe broad claims: that conflict classification, evidence-aware handling, deferral, abstention, conflict-preserving state, or asynchronous supersession alone is new.
- Still open: whether the exact combination of same-key relation classification plus an evidence-based Verify policy, compared fairly with fixed-winner and preserve-unresolved baselines under asynchronous delivery and order/convergence metrics, survives direct prior-art comparison. Novelty is not established.
- Measure the object of convergence explicitly: event set, status flags, active assertion set, conflict graph, resolved projection, and downstream retrieval can behave differently. Report admission/rejection separately from detector performance.
- Verifier risks: context/reference mismatch, unsupported evidence, prior-informed resolution, task-specific calibration, and over-abstention. Include ambiguous/unknown cases and separate classification results from policy outcomes.

Outstanding gate items

- Reconcile the five newly completed source-wide audits and the already-read CodeCRDT paper against the project's level-P standard; retrieve missing bibliography/raw artifacts where material. Continue the broader closest-threat/NLI review and active disconfirmation matrix.
- Finish the close-prior-art comparison and remaining NLI/contradiction/selective-prediction checks; reconcile ECon and related evidence tables. No independent reproduction has been performed.
- Get the authoritative exact wording/operational criteria for WH-1, WH-2 and WH-3. They remain undefined here and cannot be tested or paraphrased responsibly.
- Keep this review pack provisional. This appendix is an evidence checkpoint, not a final literature verdict, formal thesis prose, implementation authorization or proof of convergence.

Prepared-file note: these provisional audit notes now record source-wide primary-text coverage for the five required papers. The literature gate remains open; missing artifacts, broader prior-art checks and exact WH-1/2/3 wording are unresolved.